Taming the Beast!

Managing FABRIC Capacity in a CITIZEN BASED POWER BI ENVIRONMENT

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About me

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SPEAKER

Davide Lambertini

Business Intelligence Engineer @ Lucient Italia

With over a decade of experience in data field, I'm specialized in transforming data into actionable insights and bridging business and technology, mostly using Microsoft stack.
Starting in 2010 as a business analyst for a Retail company, I was a power user on tools like Excel, VBA and Access to streamline decision-making.
By 2017, I transitioned to Business Consultancy, dealing with SQL, PowerPivot, and ETL pipes.
Discovering Power BI in 2019 was love at first sight 😊, enabling me to craft impactful reports. A 2020 master's in Microsoft Big Data technologies extended my expertise to cloud platforms like Azure, Synapse, and experiment on machine learning tools like Python and R.
Now as a certified Power BI Data Analyst and Fabric Analytics Engineer in Tech Consultancy, I deliver tailored BI solutions across industries to provide fast and meaningful insights to business users.
I'm passionate about travel and exploring new cultures: they also bring a fresh perspective and opens your mind to solve complex challenges!

Microsoft CERTIFIED POWER BI DATA ANALYST ASSOCIATE

Microsoft CERTIFIED FABRIC ANALYTICS ENGINEER ASSOCIATE ASSOCIATE

Contact

davide.lambertini@lucient.it

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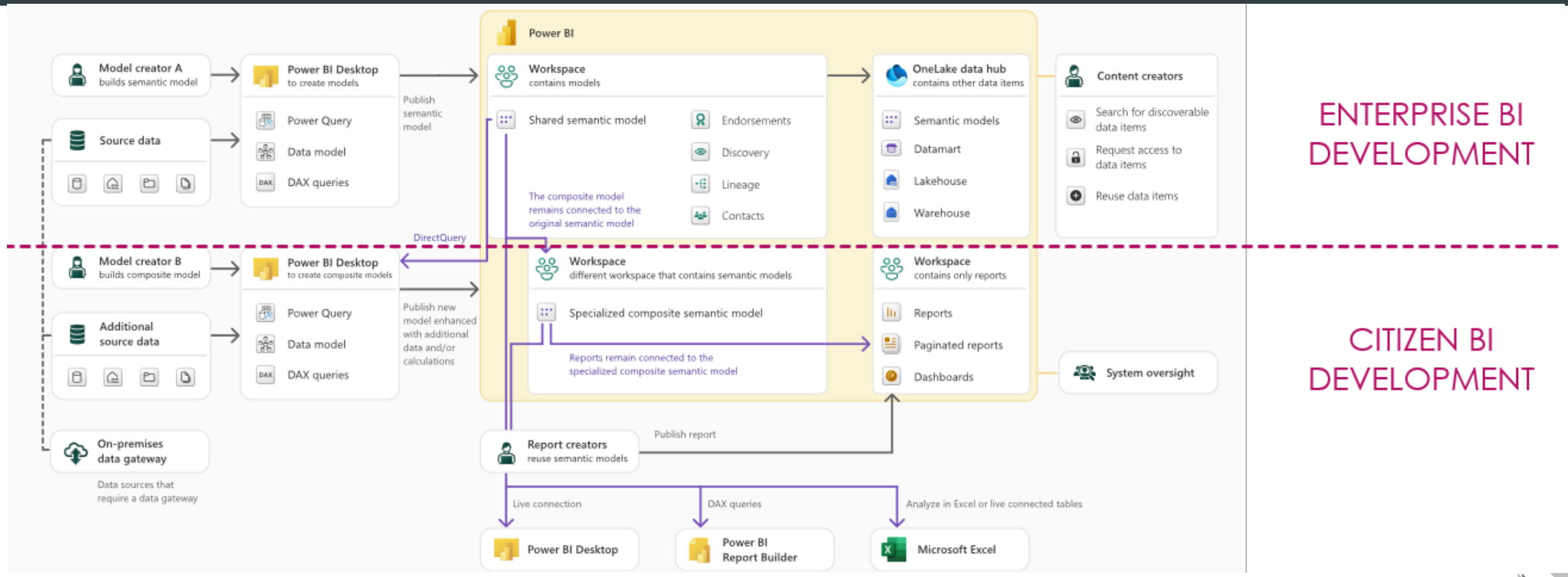
Self-Service BI Environment



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Why is Capacity so challenging?

- Resource contention, Admins cannot manage queues or kill Queries.
- Capacity has fixed size, price, and CU limits.
- Workloads are split "Background" and "Interactive" and they are treated differently: One "smooths" the other "bursts".
- Consistently "overage" of 100% capacity in 30 sec. timeframes can lead to Throttling and Rejections.

Size | P2

128

Base capacity units

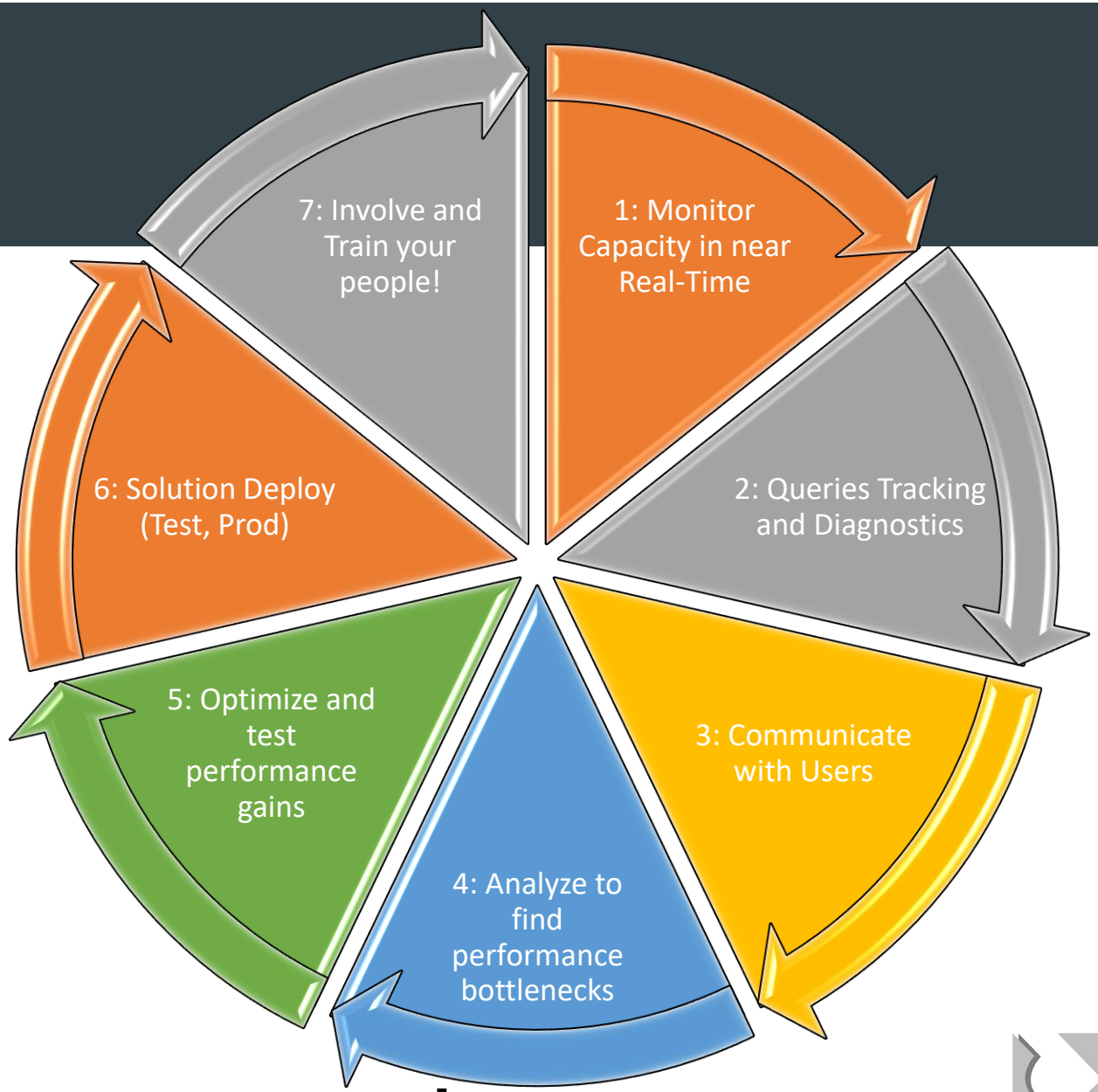
Autoscale | Off

0

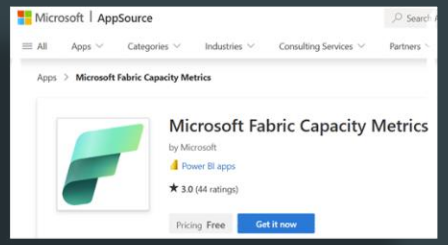
Additional capacity units in use
Max = 0

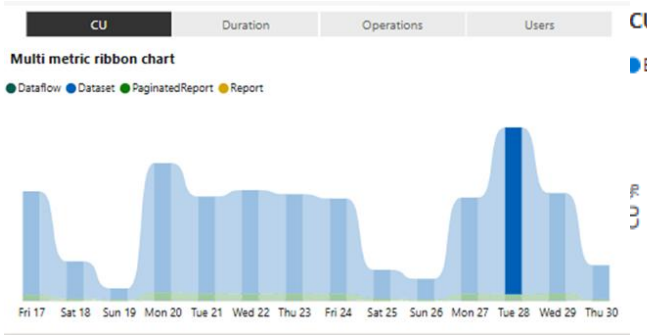
SKU	Max memory (GB)[1, 2]	Max concurrent DirectQuery connections (per semantic model)[1]	Max DirectQuery parallelism[3]	Live connection (per second)[1]	Max memory per query (GB)[1]	Model refresh parallelism	Direct Lake rows per table (in millions)[1, 4]
F2	3	5	1	2	1	1	300
F4	3	5	1	2	1	2	300
F8	3	10	1	3.75	1	5	300
F16	5	10	1	7.5	2	10	300
F32	10	10	1	15	5	20	300
F64	25	50	4 - 8	30	10	40	1,500
F128	50	75	6 - 12	60	10	80	3,000
F256	100	100	8 - 16	120	10	160	6,000
F512	200	200	10 - 20	240	20	320	12,000
F1024	400	200	12 - 24	480	40	640	24,000
F2048	400	200		960	40	1,280	24,000

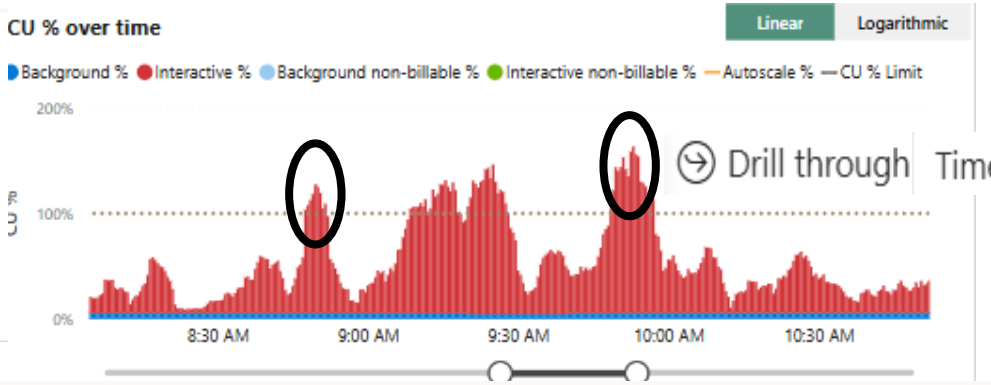
Capacity Governance Process



Near Real-Time Monitoring: Fabric Capacity Metrics App







Fabric Capacity Metrics

Start : 1/31/2025 8:49:30 AM
End : 1/31/2025 8:50:00 AM

834	2066	P2	3840
Interactive operations	Background operations	SKU	CU (s)

Interactive operations for timerange

Select optional column(s): Billing type

Workspace	Item kind	Item name	Operation	Start	End	Status	User	Duration (s)	Total CU (s)	Timepoint CU (s)	Throttling (s)	% of Base capacity	Billing type
Dataset	Dataset	M	XMLA Read ...	1/31/2025 8:34:07...	1/31/2025 8:47:13...	Failure	D ...	786	17,532.5472	1753.2547	0	45.66%	Billable
Dataset	Dataset	In	XMLA Read ...	1/31/2025 8:46:14...	1/31/2025 8:46:29...	Failure	D ...	30	1,405.2416	140.5242	0	3.66%	Billable
Dataset	Dataset	W	XMLA Read ...	1/31/2025 8:44:48...	1/31/2025 8:45:42...	Failure	n ...	54	1,227.0944	122.7094	0	3.20%	Billable
Dataset	Dataset	In	XMLA Read ...	1/31/2025 8:46:13...	1/31/2025 8:46:28...	Failure	E ...	15	788.5472	78.8547	0	2.05%	Billable
Dataset	Dataset	In	XMLA Read ...	1/31/2025 8:44:39...	1/31/2025 8:44:45...	Success	E ...	18	787.8528	78.7853	0	2.05%	Billable
Dataset	Dataset	In	XMLA Read ...	1/31/2025 8:46:14...	1/31/2025 8:46:30...	Failure	E ...	16	787.8528	78.7853	0	2.05%	Billable
Dataset	Dataset	In	XMLA Read ...	1/31/2025 8:49:01...	1/31/2025 8:49:14...	Failure	D ...	13	630.0000	63.0000	0	1.64%	Billable
Dataset	Dataset	In	XMLA Read ...	1/31/2025 8:49:08...	1/31/2025 8:49:19...	Success	D ...	11	604.8000	60.4800	0	1.58%	Billable
Dataset	Dataset	In	XMLA Read ...	1/31/2025 8:49:01...	1/31/2025 8:49:13...	Failure	D ...	12	604.4416	60.4442	0	1.57%	Billable
Dataset	Dataset	In	XMLA Read ...	1/31/2025 8:45:17...	1/31/2025 8:45:29...	Success	D ...	12	603.7472	60.3747	0	1.57%	Billable
Total								1,926	46,679.4048	4667.9405	0	121.56%	

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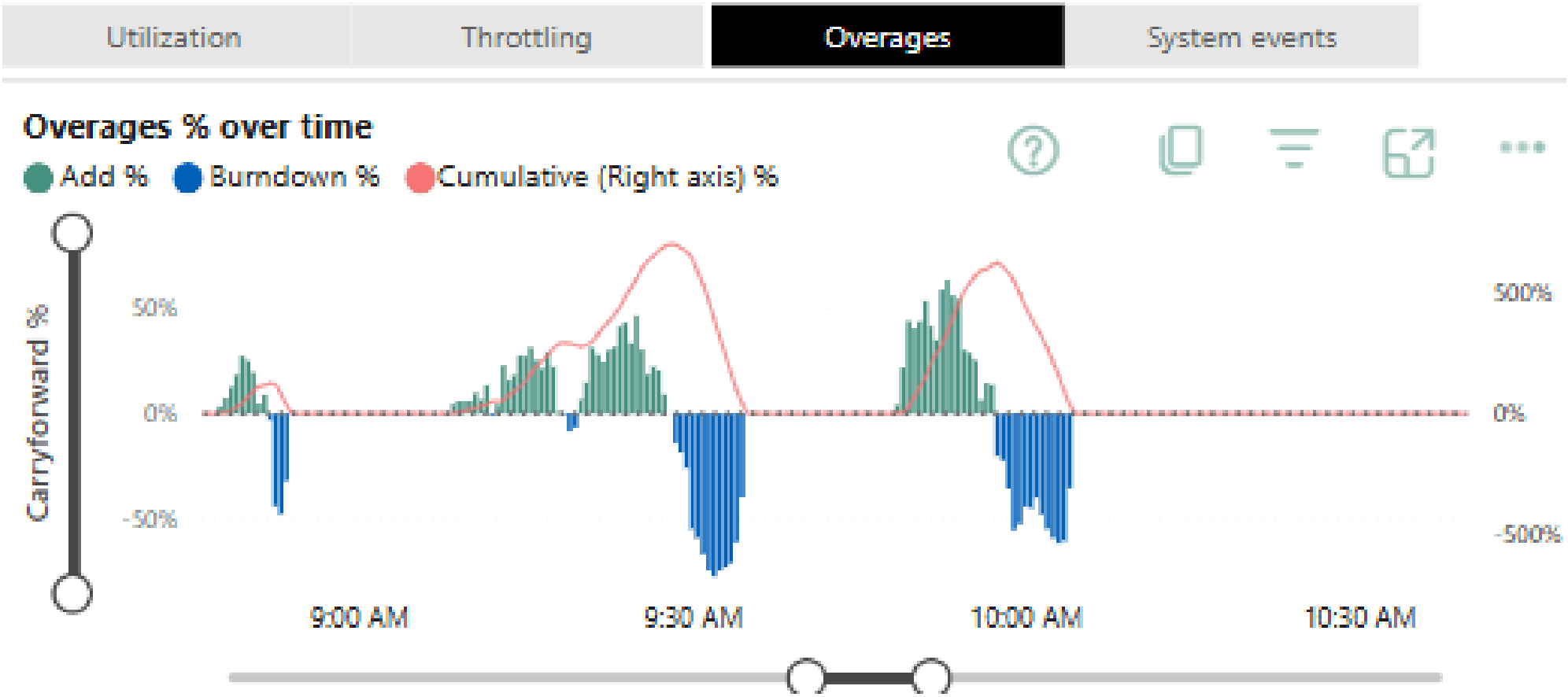
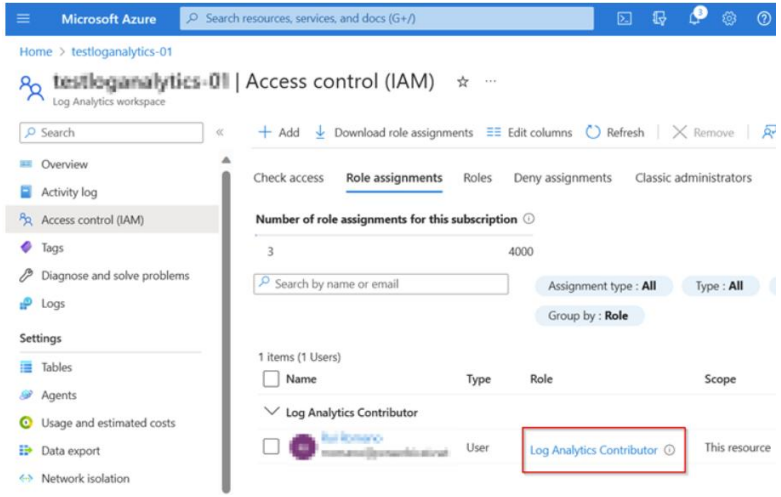
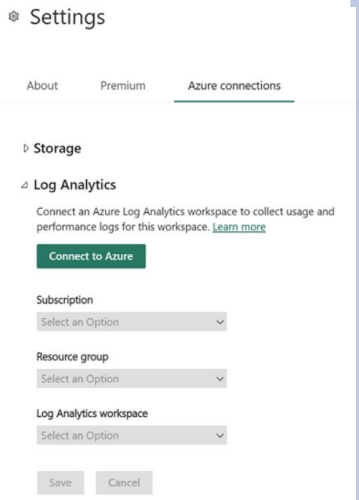
Near Real-Time Monitoring: Fabric Capacity Metrics App



Queries Tracking: Leverage Log Analytics Azure Service in a PowerBI Report!

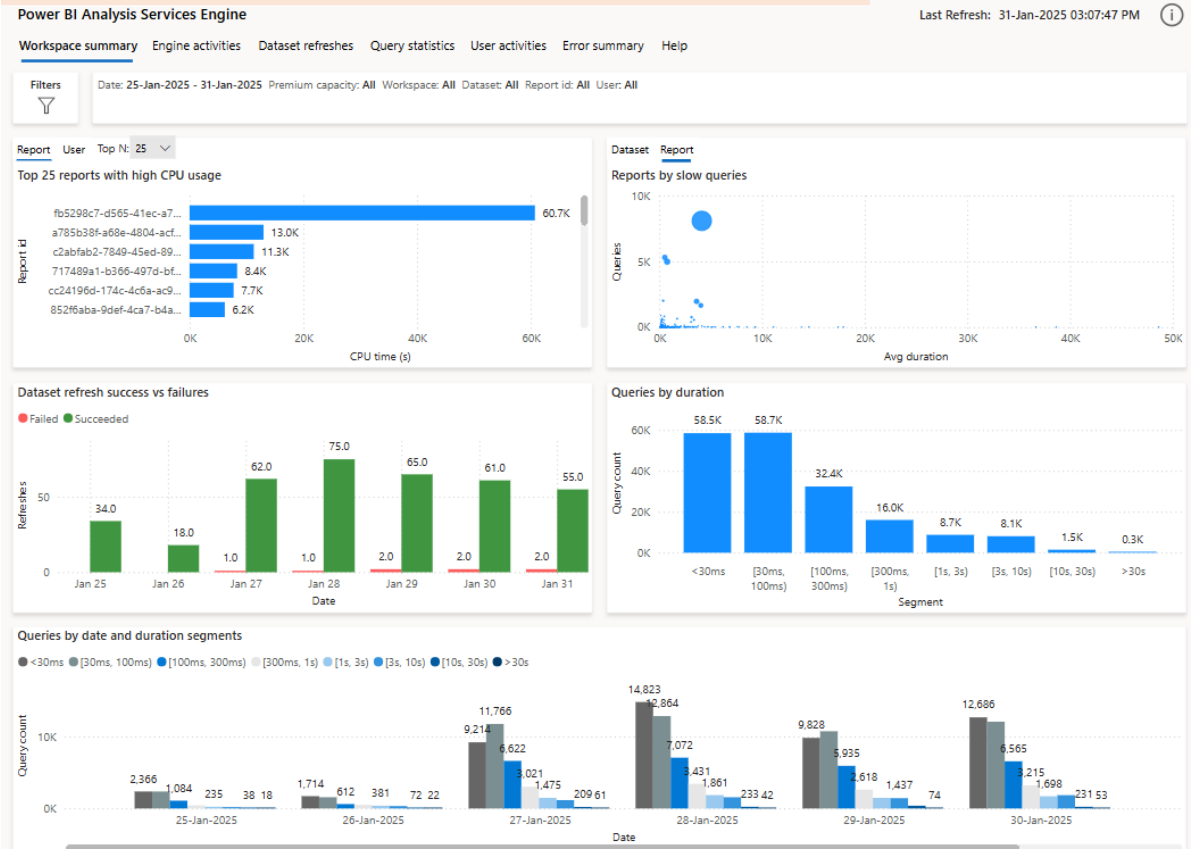




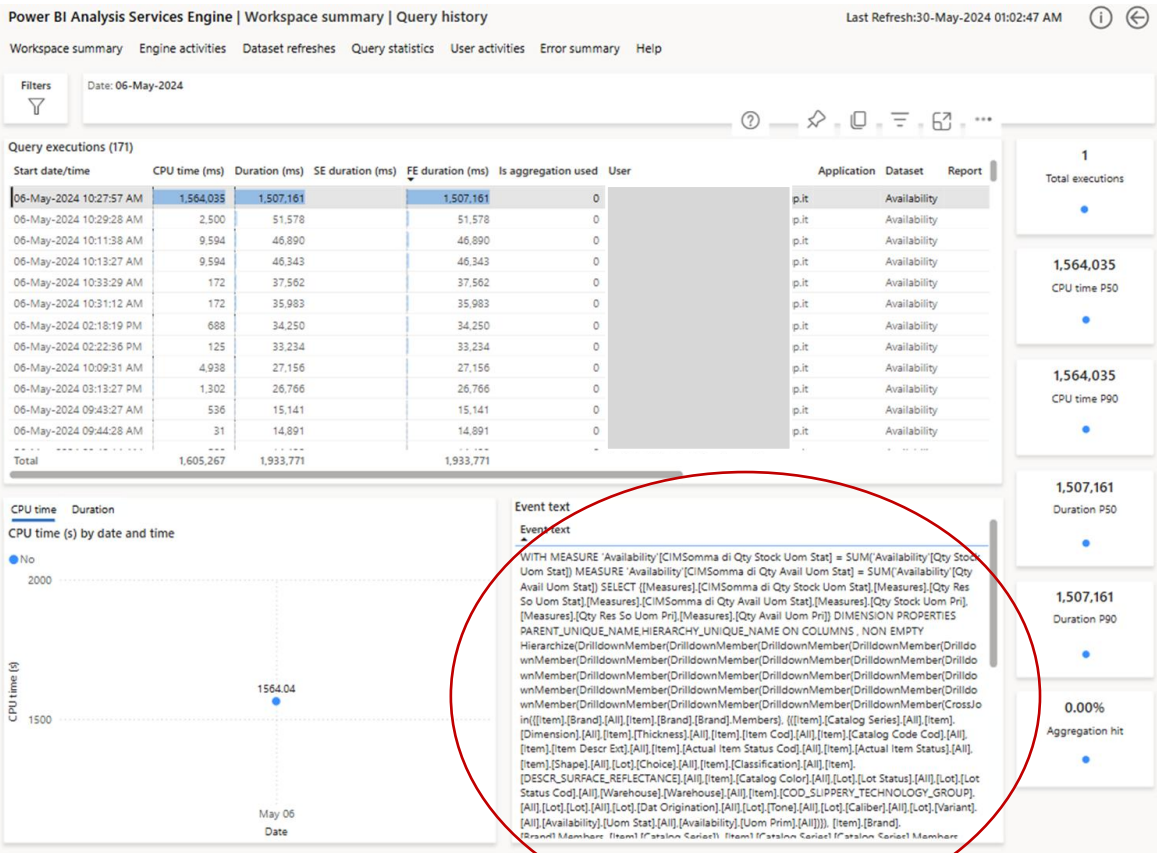
- ▶ 1: **Configure Log Analytics** and attach it to the workspace you need to monitor (full process in Microsoft docs) – Log Analytics is an Azure billable service!
- ▶2: Find **PBI Templates Open Source** for Log analytics here: [GitHub - microsoft/PowerBI-LogAnalytics-Template-Reports: The repository hosts Power BI template files that can be used to analyze Power BI data after you send it to Azure Log Analytics.]
- ▶ 3: **Set Parameters, Refresh, Publish** and schedule into your Workspace

Queries Tracking: Log Analytics PBI Template

From Dashboards..



Drill Though to Detail..



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Queries Tracking: Log Analytics in Near Real-Time (KQL)

Home > LogAnalyticsSASS

LogAnalyticsSASS | Logs

Log Analytics workspace

New Query 1* | New Query 2* | New Query 3* | **New Query 4*** | Save | Share | Queries hub

Run | Time range : **Set in query** | Show : **1000 results** | KQL mode

```
    CpuTimeMs,
    DurationMs,
    ExecutingUser,
    DurationSec = DurationMs / 1000.0
| join kind=leftouter executionMetrics on XmlaRequestId
//| join kind=leftouter MemStats
| extend MaxPeakMemoryGB, // Convert bytes to GB and round
        AvgPeakMem = toscalar(MemStats | project AvgPeakMem),
        StdPeakMem = toscalar(MemStats | project StdPeakMem)
| extend AnomalyScore = (MaxPeakMemoryGB - AvgPeakMem) / StdPeakMem  // Calculate z-score for duration
| where AnomalyScore >= AnomalyThreshold  // Filter to only anomalously heavy queries
| order by AnomalyScore desc  // Order by heaviest (RAM)
```

Results | Chart

	CpuTimeMs	Dura...	XmlaRequestId1	MaxPeakMem...	AvgPeak...	StdPeak...	Ano...
	68219	68.501	afd56983-35e9-4c58-...	10	0.01384167...	0.22625557...	44.13662957...
	76328	76.297	1ea8eac4-ab26-4d88...	10	0.01384167...	0.22625557...	44.13662957...
[IsDM5Total] DESC, 'Shipped'[udc], [IsDM7Total] DESC, 'Item Shipped'[Item], [IsDM9Total] DESC,...	75625	72.094	77996cae-5775-4b60...	10	0.01384167...	0.22625557...	44.13662957...
Warehouse Descr], [IsDM1Total] DESC, 'Shipped'[Customer Descr], [IsDM3Total] DESC, 'Shipped'...	72188	69.203	049ae552-24b7-4aaf-...	9.33	0.01384167...	0.22625557...	41.17537652...
	56953	7.328	7cfb9ff1-cd8f-439e-b...	7.06	0.01384167...	0.22625557...	31.14247439...

Columns

1s 800ms | Display time (UTC+00:00) | Query details | 1 - 6 of 111

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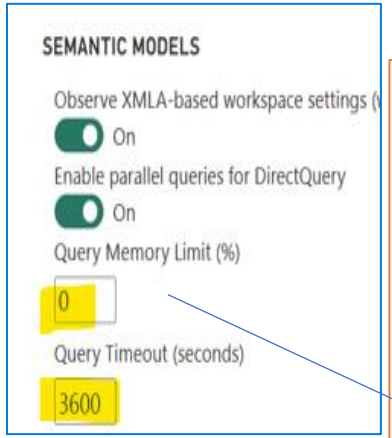
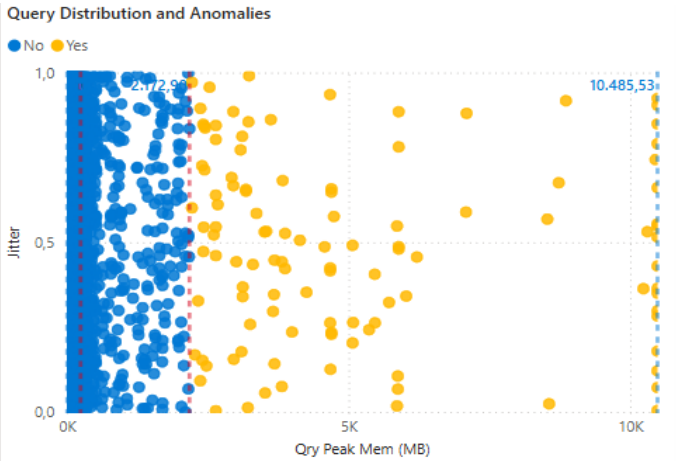
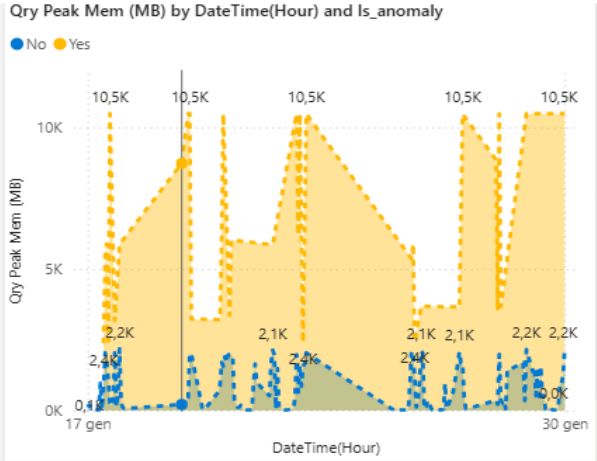
Anomaly detection:
What is the best fit for CU limit?

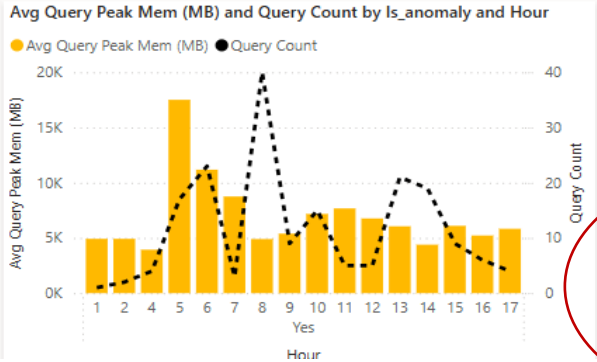


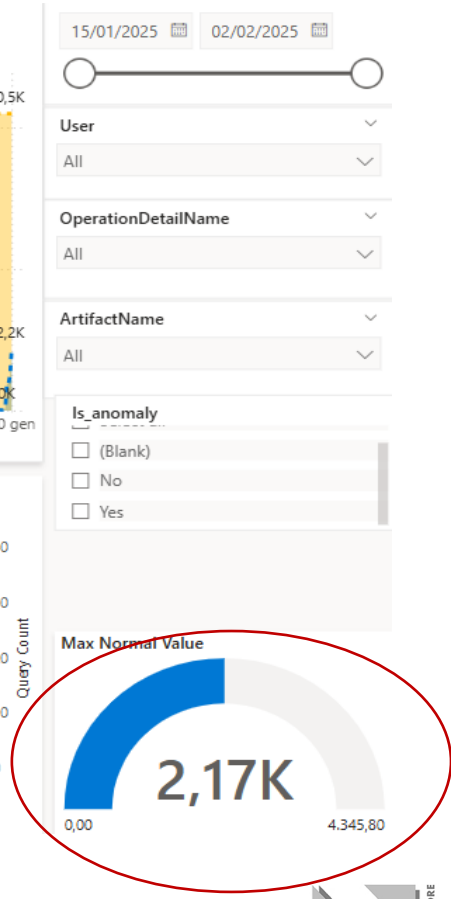
SKU	Automatic Query Memory Limit
F2	1 GB
F4	1 GB
F8/EM1/A1	1 GB
F16/EM2/A2	2 GB
F32/EM3/A3	5 GB
F64/P1/A4	10 GB
F128/P2/A5	10 GB



Is_anomaly	Query Count	%GT Query Count	Query Total Memory (MB)	%GT Query Total Memory (MB)	Qry Peak Mem (MB)
Yes	183	0,18%	578.579,85	21,76%	10.485,53
No	102280	99,82%	2.080.709,00	78,24%	2.172,90
Total	102463	100,00%	2.659.288,84	100,00%	10.485,53







Capacity Management: Theory vs. Practice

THEORY

- Refreshes are smoothed background ops so they are not so critical...
- Log Analytics tracks all queries and I can see where they lands and which report and user is issuing the query..
- Setting specific Memory per query limits or timeouts to the capacity, will avoid anomalous queries to block my capacity..

PRACTICE

- ..Downstream Semantic model Refreshes are seen as interactive ops.
- ..Log Analytics tracks query landing but if user is using local tools, you cannot know which is the report.
- ..If you set per Query limits the query is issued anyway, it just fails before when it hits the limits. It just helps..

Capacity Overhead: So…What to do?

		+	−
SCALE UP	Increase Capacity Size Or Enable AutoScale	Fast & Simple: Increase capacity Size	It doubles costs, Autoscale is min 24 hours
SCALE OUT	Split Workflows to different Capacities	No costs in mixing a Shared Capacity + a Premium Capacity	Needs Strategical decisions and to consider many limits
OPTIMIZE	Find Queries that mostly burns your capacity and optimize them	Impactful, it can avoid to surge problems	Difficult to involve users, Time consuming and skills needed

DATA SATURDAYS

20+ OPTIMIZATION STRATEGIES :

1. ETL Phase
2. Data Models
3. Visuals & Dax
4. Composite

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TAME ETL&IMPORTS

1. ENABLE QUERY FOLDING & ENSURE EARLY FILTERING WHILE FOLDING HAPPENS
2. REDUCE VOLUMES; COLUMNS, TABLES OR AGGREGATE ROWS
3. IMPROVE DATA COMPRESSION USING OPTIMAL DATA TYPES
4. USE INCREMENTAL REFRESH INSTEAD OF FULL DATASET LOADS
5. MAINTAIN CLEAN AND COMPACT POWER QUERY STEPS
6. TRANSLATE MDX QUERIES IN DAX NATIVE QUERIES
7. AVOID CARTESIAN PRODUCTS WHILE DEVELOPING M

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Optimization ETL Example
Query Folding & Steps

Chaotic and steps redundant

Query Folding broken here!

APPLIED STEPS

- Source
- Navigation
- Added Items
- Added Items1
- Added Items2
- Changed Type
- Filtered Rows
- Added Items3
- Added Items4
- Additional Column

Edit Settings
Rename
Delete
Delete Until End
Insert Step After
Move Up
Move Down
Extract Previous
View Native Query
Properties...

Late Filtering, after query folding is broken.Bad Performances

Not supported by query folding.

Duration (s)	Total CU (s)	Timepoint CU (s)	Throttling (s)	% of Base capacity
888	17,041.8528	1704.1853	0	44.38%

Clear and Compact

APPLIED STEPS

- Source
- Navigation
- Added Items
- Filtered Rows
- Changed Type
- Additional Column

All Add Items in one step

Early Filtering, while query folding is occuring

2 Steps that doesn't fold moved to the end!

Duration (s)	Total CU (s)	Timepoint CU (s)	Throttling (s)	% of Base capacity	Billing type
38	984.5472	98.4547	0	2.56%	Billable

Optimization ETL Example
Dax Native Q. & Early Filtering

APPLIED STEPS

- Origine
- Navigation
- Elementi aggiunti
- Filtrate righe
- Elementi aggiunti1
- Filtrate righe1
- Elementi aggiunti2
- Filtrate righe3
- Elementi aggiunti3
- Filtrate righe4
- Elementi aggiunti4
- Elementi aggiunti5
- ~~Elementi aggiunti6~~
- Colonna Condiz. Account Cod
- Filtrate righe2
- Rimosse colonne
- Inner Join Account Cod
- Rinominate colonne

Duration (s)	Total CU (s)	Timepoint CU (s)	Throttling (s)	% of Base capacity
375	8,537.2000	853.7200	0	22.23%
353	8,058.0416	805.8042	0	20.98%
332	7,551.2416	755.1242	0	19.66%
322	7,378.0000	737.8000	0	19.21%
247	5,595.8112	559.5811	0	14.57%

```
= "EVALUATE " &
EVALUATE SUMMARIZECOLUMNS('Company'[Company], 'Company'[Legal Entity], 'Account ORS'[Costs Level 1], 'Account ORS'[Costs Level 2], 'Account ORS'[Costs Level
3], 'Time 544'[Year 544], 'Time 544'[Month 544], 'Time 544'[YearMonth 544], 'Item'[Item Cod], 'Item'[Item], 'Supplier'[Supplier
Cod], 'Supplier'[Supplier], 'Project'[Project Cod], 'Project'[Project], 'Cost Center'[Cost Center Cod], 'Account'[Account Cod], 'Journal Entry'[Journal Entry], FILTER
( 'Account', 'Account'[Account Cod] IN
{ "8A01090001", "8V01090001", "8N01090001", "8O01090001", "8A01090002", "8V01090002", "8N01090002", "8O01090002", "8A01090003", "8V01090003", "8N010
90003", "8O01090003", "8A01090004", "8V01090004", "8N01090004", "8O01090004", "8A01090005", "8V01090005", "8N01090005", "8O01090005", "8A04050004", "
8V04050004", "8A01090006", "8V01090006", "8N01090006", "8O01090006", "8A01090007", "8V01090007", "8N01090007", "8O01090007", "8A01090008", "8V01090
008", "8N01090008", "8O01090008", "8A01090009", "8V01090009", "8N01090009", "8O01090009", "8A01090012", "8V01090012", "8N01090012", "8O01090012", "8A
01090013", "8V01090013", "8N01090013", "8O01090013", "8A01090019", "8V01090019", "8N01090019", "8O01090019", "8A01090020", "8V01090020", "8N0109002
0", "8O01090020", "8A04050011", "8V04050011", "8N04050011", "8O04050011", "8A01090021", "8V01090021", "8N01090021", "8O01090021", "8A02010004", "8A02
( 'Company', 'Company'[Legal Entity] IN { "014", "G66", "K60", "K61", "N04", "U04", "U14", "U66" } ), FILTER( 'Account ORS', 'Account ORS'[Costs Level 2] IN
{ "Conversion_Costs", "Missing Hierarchy" } ), FILTER( 'Account ORS', 'Account ORS'[Costs Level 3] IN { "Consumable_Mfg", "Maintenance_Mfg", "Missing Hierarchy" } ),
```

Query Settin

PROPERTI

APPLIED STEPS

- Map_accounts_List
- Map_accounts_List_Text
- DAXQuery
- Source
- Renamed Columns
- Colonna condizionale aggiunta
- Filtrate righe
- Rimosse colonne
- Rinominate colonne

Duration (s)	Total CU (s)	Timepoint CU (s)	Throttling (s)	% of Base capacity
17	387.4528	38.7453	0	1.01%
5	150.5056	15.0506	0	0.39%
4	108.1472	10.8147	0	0.28%
1	60.9056	6.0906	0	0.16%
2	52.1472	5.2147	0	0.14%
2	50.4000	5.0400	0	0.13%

Refresh Time From 30 min to 30 seconds!

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TAME DATA MODELS

8. WARN PEOPLE ON NEVER PUBLISH REPORTS OR MODELS WITH SLOW VISUALS/SLOW RUNNING REFRESHES, INSTRUCT TO RECOGNIZE THEM
9. PREFER STAR SCHEMA AND BEWARE OF DENORMALIZATION
10. MINIMIZE USAGE OF CALCULATED COLUMNS
11. ENSURE LOGICAL FILTER PROPAGATION AND ARRANGE YOUR MODEL LOGICALLY
12. USE EXPLICIT MEASURES TO IMPROVE FILTER BEHAVIOR
13. LIMIT THE USE OF M2M AND BI-DIRECTIONAL RELATIONSHIPS TO NECESSARY SCENARIOS.
14. DISABLE AUTO DATETIME & USE A SINGLE CALENDAR TABLE

Optimization Modeling Example
Better compression = lighter models!

Hey, Data Types matters!

Name	Cardinality	Total Size	Data	% Table
MOV MAGAUT...	**103.691**	**11.712.683**	**762.528**	
DataCreazioneL...	103.677	6.153.592	276.640	52,54%
IdentificativoLas...	103.691	4.794.298	276.640	40,93%
CodiceUds	13.865	758.993	207.296	6,48%
UdsCurrentQua...	45	2.788	928	0,02%
UdsTargetQuant...	45	2.756	896	0,02%
MOV MAGAUT...	**103.691**	**8.795.931**	**785.672**	
IdentificativoLas...	103.691	4.794.298	276.640	54,51%
DataCreazioneL...	51.566	3.131.704	207.512	35,60%
CodiceUds	13.865	758.993	207.296	8,63%
DataCreazioneL...	275	105.136	92.272	1,20%
UdsTargetQuant...	45	2.820	960	0,03%
UdsCurrentQua...	45	2.724	864	0,03%

Beware of Autodatetime!

Name	Cardinality	Total Size	Data	Dictionary	% DB
(R*) Resources	8.360.368	560.347.879	247.308.552	182.635.599	27,75%
(R) Activities	2.435.111	451.420.546	53.617.752	295.075.034	22,36%
(M*) Batch Materials	2.026.598	264.978.100	89.230.840	120.747.420	13,12%
(M) Item	481.851	186.123.354	36.275.688	127.247.178	9,22%
LocalDateTable_c4b...	2.737.127	170.282.420	12.600.320	89.892.116	8,43%
Steps	1.834.676	160.143.376	46.445.728	79.304.744	7,93%
Resources Primary	1.810.520	86.966.989	18.955.568	40.967.845	4,31%
Batches	219.872	64.088.273	6.985.176	43.809.025	3,17%
Batch Formula	313.296	35.837.426	3.450.240	25.655.290	1,77%
Batch Groups	36.089	15.061.211	983.344	11.185.099	0,75%
LocalDateTable_c42...	65.013	5.031.028	257.496	3.162.764	0,25%
LocalDateTable_256...	28.489	2.338.264	114.360	1.517.712	0,12%
LocalDateTable_309...	8.035	633.056	32.448	400.992	0,03%
LocalDateTable_202...	8.035	633.056	32.448	400.992	0,03%
LocalDateTable_8c1...	8.035	633.056	32.448	400.992	0,03%
LocalDateTable_040...	8.035	633.056	32.448	400.992	0,03%
LocalDateTable_ec2...	8.035	633.056	32.448	400.992	0,03%
LocalDateTable_987...	8.035	633.056	32.448	400.992	0,03%
LocalDateTable_988...	7.670	617.596	31.112	395.908	0,03%
LocalDateTable_a31...	7.670	617.596	31.112	395.908	0,03%
LocalDateTable_5cf8...	7.670	617.596	31.112	395.908	0,03%
LocalDateTable_766...	7.670	617.596	31.112	395.908	0,03%
LocalDateTable_dbf...	6.574	571.196	27.080	380.660	0,03%
LocalDateTable_a16...	6.574	571.196	27.080	380.660	0,03%

Too Much Calc Cols & denormalization!

Query executions (607)

Start date/time	CPU time (ms)	Duration (ms)	SE duration (ms)	FE duration (ms)	Is aggregation used	User	Application	Dataset	Report	Premium capacity
29-Apr-2024 07:42:55 PM	3,640,656	3,600,758		3,600,758	0	i@m...	Power BI Desktop966a46e3-b...			2F2DA2F1-09A1-4
29-Apr-2024 05:14:02 PM	3,593,923	3,553,387		3,553,387	0	i@m...	Power BI Desktop28ed48fb-5...			2F2DA2F1-09A1-4
29-Apr-2024 04:05:02 PM	2,323,156	2,283,246		2,283,246	0	i@m...	Power BI Desktop67365659-6...			2F2DA2F1-09A1-4
29-Apr-2024 06:10:43 PM	1,205,580	1,166,327		1,166,327	0	i@m...	Power BI Desktope288ef55-c...			2F2DA2F1-09A1-4
29-Apr-2024 07:36:18 AM	684,250	152,751		152,751	0	i@m...	Power BI Desktop1d87ce0f-6...			2F2DA2F1-09A1-4
29-Apr-2024 07:36:15 AM	684,126	150,486		150,486	0	i@m...	Power BI Desktop1d87ce0f-6...			2F2DA2F1-09A1-4
29-Apr-2024 05:48:48 PM	674,688	147,705		147,705	0	i@m...	Power BI Desktope288ef55-c...			2F2DA2F1-09A1-4
29-Apr-2024 05:48:50 PM	672,938	150,049		150,049	0	i@m...	Power BI Desktope288ef55-c...			2F2DA2F1-09A1-4
29-Apr-2024 03:59:44 PM	667,203	145,955		145,955	0	i@m...	Power BI Desktop28ed48fb-5...			2F2DA2F1-09A1-4
29-Apr-2024 06:40:22 PM	655,625	147,658		147,658	0	i@m...	Power BI Desktop966a46e3-b...			2F2DA2F1-09A1-4
29-Apr-2024 05:21:18 PM	655,609	142,673		142,673	0	i@m...	Power BI Desktop74f1780d-e...			2F2DA2F1-09A1-4
Total	24,851,841	14,271,477		14,271,477						

CPU time | Duration

CPU time (s) by date and time

Event text

Event text

Define COLUMN 'Orders'[ASDQ_Shipping date (REF)] = IF(Orders[Delivery Creation Date]>Orders[ASDQ_shipping date],Orders[Delivery Creation Date],Orders[ASDQ_shipping date]) COLUMN 'Orders'[ASDQ_WHS LT] = CONVERT(Orders[ASDQ_Scheduled ship date (REF)]-Orders[Delivery Creation Date], Double) COLUMN 'Orders'[ASDQ_Scheduled ship date (REF)] = IF(Orders[Schedule Ship Date]<Orders[Delivery Creation Date],Orders[Delivery Creation Date],IF(Orders[Schedule Ship Date]>IF(Orders[Delivery Creation Date]>Orders[ASDQ_shipping date],Orders[Delivery Creation Date],Orders[ASDQ_shipping date]),IF(Orders[Delivery Creation Date]>Orders[ASDQ_shipping date],Orders[Delivery Creation Date],Orders[ASDQ_shipping date]),Orders[Schedule Ship Date])) COLUMN 'Orders'[ASDQ_Delivery LT] = CONVERT(Orders[Delivery Creation Date]-Orders[ASDQ_Reservation date (DEF)], Double) COLUMN 'Orders'[ASDQ_shipping date] = related('Time Shipping'[Day]) COLUMN 'Orders'[ASDQ_Availability LT] = if(Orders[ASDQ_Reservation date (DEF)]-Orders[ASDQ_max Creation date line] <0,0,Orders[ASDQ_Reservation date (DEF)]-Orders[ASDQ_max Creation date line]) COLUMN 'Orders'[ASDQ_Reservation date (DEF)] = IF(Orders[ASDQ_MIN MAX RESERVATION/LINE CREATION]>Orders[Delivery Creation Date],Orders[Delivery Creation Date],Orders[ASDQ_MIN MAX RESERVATION/LINE CREATION]) COLUMN 'Orders'[ASDQ_MIN MAX RESERVATION/LINE CREATION] = IF(Orders[ASDQ_MAX Reservation date/Promise (MOD)]<Orders[ASDQ_max Creation date line],Orders[ASDQ_max Creation date line],Orders[ASDQ_MAX Reservation

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DATA SATURDAYS

TAME VISUALS & DAX

15. USE FILTER ARGUMENTS EFFICIENTLY IN DAX: FILTER COLUMNS, NOT ENTIRE TABLES
16. OPTIMIZE DISTINCTCOUNT USAGE ON HIGH-CARDINALITY COLUMNS
17. PREFER VARIABLES OVER NESTED ITERATORS IN DAX
18. USE DIVIDE INSTEAD OF IF(/) PATTERNS FOR SAFE CALCULATIONS
19. OPTIMIZE VISUALS PER REPORT PAGE
20. UTILIZE PERFORMANCE ANALYZER FOR FRONT-END OPTIMIZATION
21. USE PAUSE VISUALS FEATURE WHILE DEVELOPING IN DIRECTQUERY MODE

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DATA SATURDAYS

Imagine one Thin Report in Direct Query.
A Visual is not rendering and it's Running out of memory!

Error fetching data for this visual See details

Error fetching data for this visual

OLE DB or ODBC error: Resource Governing: This query uses more memory than the configured limit. The query — or calculations referenced by it — might be too memory-intensive to run. Either simplify the query or its calculations, or if using Power BI Premium, you may reach out to your capacity administrator to see if they can increase the per-query memory limit. More details: consumed memory 10547 MB, memory limit 10240 MB. See https://go.microsoft.com/fwlink/?linkid=2159752 to learn more.

Dettagli tecnici:
RootActivityId: 441bb918-8e39-4484-8a9f-33473e819783
Data (UTC): 2/9/2025 2:32:51 PM.

```
Num_SKU_B/O =
    CALCULATE(
        DISTINCTCOUNT('Item'[Catalog Code]),
        FILTER('Portfolio Trend',[Quantity Back Order]>0)
        )
```

Mmh, seen a single measure is fully responsible of the perf issue on this Visual.
Let's Rewrite it:

Performance analyzer

Start recor... Refresh vis... St...

Clear Export

Name	Duration (ms)
Refreshed visual	-
Button	88
Slicer	85
Table	3006
DAX query	2804
Direct query	2753
Visual display	74
Other	128
Copy query	
Run in DAX Query View	
Table	1454

```
Num_SKU_B/O opt =
CALCULATE(
        SUMX(DISTINCT('Item'[Catalog Code]), 1),
        --alternative to distinctcount
        FILTER(VALUES('Portfolio Trend'[Order Number]), [Quantity Back Order] > 0) ,
        --filter order number with backorder >0
        CROSSFILTER('Item'[SUR_ART], 'Portfolio Trend'[SUR_ART], Both)
        --enable crossfilter at runtime to propagate relationship
    )
```

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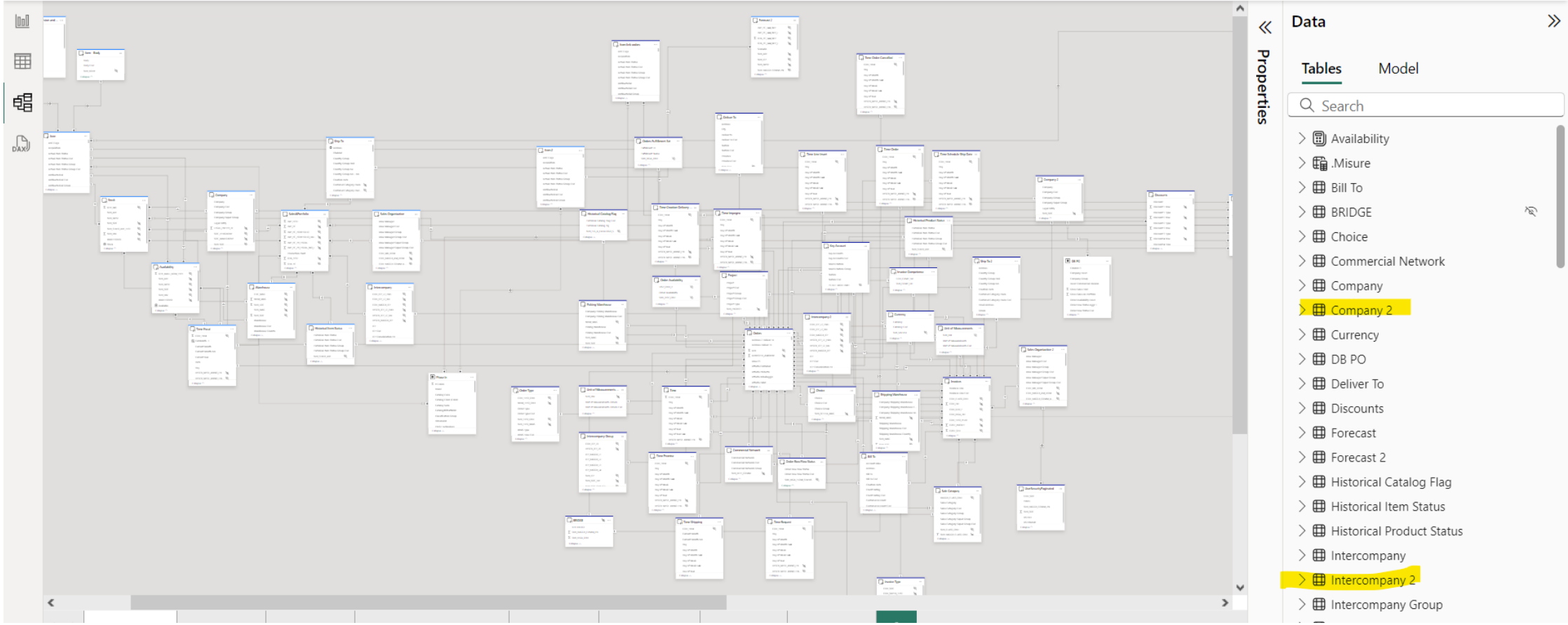
TAME COMPOSITES

22. LEVERAGE AGGREGATIONS IN COMPOSITE MODELS
23. AVOID BUSINESS KEY JOINS USING CALCULATED COLUMNS FOR COMPOSITE MODELS
24. ELIMINATE REDUNDANT DIMENSIONS IN COMPOSITE MODELS
25. ARRANGE TABLES LOGICALLY IN MODEL VIEW
26. USE DUAL MODE IN SQL SERVER

Composite models Mistakes Example
..how to control filter propagation?



DATA SATURDAYS

Performance Optimization Toolbox

Microsoft SQL Server Profiler

DAX STUDIO

Tabular Editor

VertiPaq Analyzer

Bravo v1.0 FOR POWER BI

4 LESSONS LEARNED:

1. Optimization is a journey, not a destination.
2. Foster a data-driven culture focusing on performance and scalability.
3. Empower skilled Power Users for a strong organizational foundation.
4. Power BI is powerful but requires strong governance - start right to avoid future chaos.

DATA SATURDAYS

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THANKS!

DATA SATURDAYS

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